ShuffleSnap: strided windowed exact reassignment for point-to-grid layout

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Abstract

Giving each point of a 2D embedding its own grid cell with the least total squared displacement is a linear assignment problem too large for dense solvers at 10^5 points. ShuffleSnap, inspired by Shellsort, solves small exact assignment problems in windows whose cells lie a stride apart, and shrinks the stride to one. Against certified optima it comes within hundredths of a percent on clustered layouts, including real UMAP embeddings, in seconds. Its errors are loops of one-cell shifts that no window can see, which make uniform points, concave masks and surplus cells hard.

1 From points to cells

An image collection is often shown by embedding the items in the plane, for example with t-SNE or UMAP [6, 10], and placing each thumbnail in its own grid cell. Rounding every point to its nearest cell causes collisions and holes, so the cells must be assigned jointly: we seek the assignment σ of points p_i to distinct cells $c_{\sigma(i)}$ that minimizes $\sum_i \|p_i - c_{\sigma(i)}\|^2$. This linear assignment problem, also a discrete optimal-transport problem [8], is solved exactly by dense solvers [2, 4] that need the full cost matrix: 80 GB at 10^5 points. Practical tools instead split the cloud recursively, like RasterFairy [3], or merge randomized partitions, like BSP-OT [1].

ShuffleSnap borrows from Shellsort [9], which sorts interleaved subsequences at a shrinking gap. It solves only small assignment problems, but exactly, with their cells spread across the grid first and brought together later (Figure 1). We treat the idea as a hypothesis, and measure its true gaps with a solver that proves optimality on the same instances.

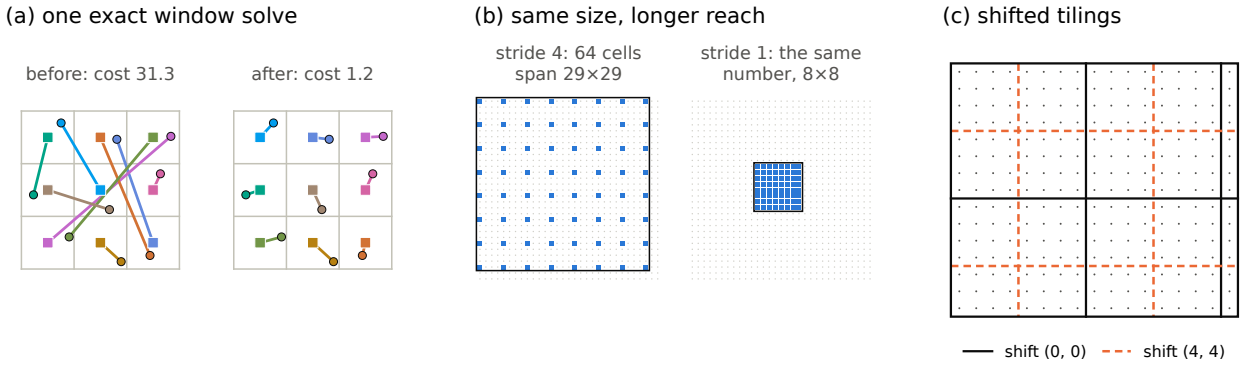


Figure 1: (a) A window collects the points that occupy its cells and reassigns them among the same cells by an exact linear assignment. (b) At stride s a window takes every s-th row and column: it keeps 64 cells but spans $7s + 1$ cells per axis, so points can travel far. Strides shrink from left to right. (c) Each subgrid is tiled four times, shifted by half a window, so that points can cross every window border.

2 Coarse to fine

ShuffleSnap starts from a random assignment. At stride s we keep every s-th row and every s-th column of the grid, which splits it into s^2 interleaved subgrids, one for each choice of first row and first column. Each subgrid is cut into windows of 8×8 of its cells. In every window, the current occupants are reassigned among the window's cells by solving a linear assignment problem of at most 64×64 exactly, with costs taken from the original point coordinates. The new assignment replaces the old one only if it lowers the window's cost by more than a relative 10^{-12}, so the layout stays valid, its cost never increases, and ties cannot cause cycles.

A round tiles every subgrid four times, shifted by $(0,0)$, $(0,4)$, $(4,0)$ and $(4,4)$. Windows that cross an edge are clipped, not dropped. The windows of one tiling are disjoint, so they are solved in parallel. On an axis of length L the first stride is the smallest power of two with $8s \geq L$, which makes the first windows span the grid. After each round the stride is divided by $\sqrt{2}$ and rounded; at stride 1, rounds repeat until one moves no point. Each level costs about the same, linear in the number of cells, and the no-move stop is a fixed point of the windows, not a proof of optimality. We chose the ratio $\sqrt{2}$ experimentally: no ratio between 1.1 and 3 was clearly better. A plausible reason is that it halves the area a window spans from one level to the next, rather than quartering it, so each level starts from a layout already sorted at nearly its own scale. Larger windows were better in every test but slower; besides the default 8×8 we offer 5×5 (*fast*) and 12×12 (*quality*).

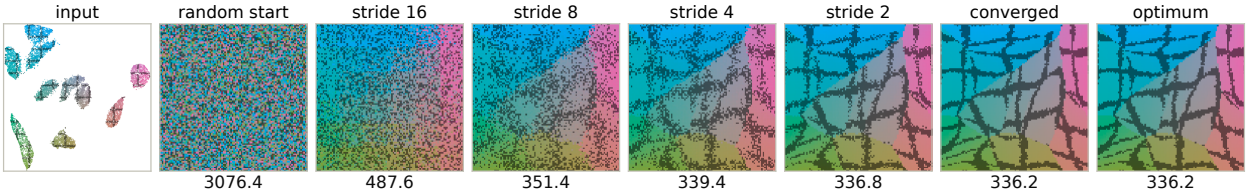


Figure 2: Progress on 10,000 points of a UMAP embedding of MNIST digits [5] (every second stride shown). Every point keeps the color of its input position, and the dark bands are graph-paper lines drawn on the input: both travel with the points. Numbers are the mean squared displacement in cells^2.

Targets and implementation. A target can be any set of cells: a rectangle, a disk, the shape of a letter. Each axis has its own stride, masked cells never enter a window, and empty cells do, so with more cells than points the solver also chooses which cells stay empty. The kernel is C++; a 70-line NumPy version reproduces it exactly.

Certified optima. Measuring gaps needs the true optimum, but at 10^5 points a dense solver does not fit in memory. We solve a min-cost flow with OR-Tools [7] on sparse candidate edges: each point's 12 nearest cells and the neighbourhood of its cell in a valid starting layout (ShuffleSnap's). As this may miss an edge, we check it against the full, dense problem, whose optimal cost any cell potentials v bound from below:

$$\mathrm{LB}(v) = \sum_i \min_j \big(\|p_i - c_j\|^2 - v_j \big) + \sum_j v_j - (M - N) \max_j v_j,$$

where $M \geq N$ is the number of cells. We derive v from the sparse solution, evaluate the bound over *all* cells, add every edge it shows to be missing, and repeat until the bound meets the cost. Then no assignment of the dense problem is cheaper: the result is its exact optimum; the start affects only the time. Where it fits, SciPy's dense solver found the same optimum on 600 instances of up to 2,500 points and on all ten distributions at 10^4 points (relative difference below 10^{-9}). It certified every benchmark instance (up to 2×10^5 points); all gaps below use it.

3 Evidence and limits

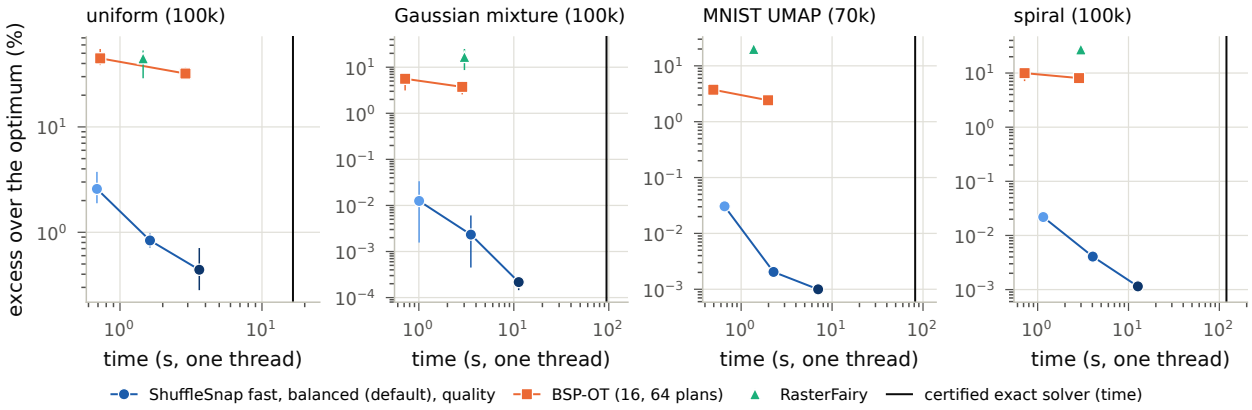


Figure 3: Excess over the certified optimum against time on one core. Points are means over three instances (for MNIST, three runs on the one embedding); bars span their range. The three ShuffleSnap presets use 5×5, 8×8 (default) and 12×12 windows from left to right. The vertical line is the time of the certified exact solver.

We ran ten synthetic distributions with 10^5 points and the complete UMAP embeddings of MNIST and Fashion-MNIST [11] (70,000 points each) on one workstation. Every method received identical coordinates and cells, and all times include each method's own preparation. Figure 3 shows three of them and MNIST.

On clustered data, the kind of layout that embeddings produce, ShuffleSnap is close to optimal. It is within 0.06% of the optimum on nine of the ten distributions and takes a median of 2.3 s on one core, or 0.19 s on 24 threads. On the two real embeddings it is within 0.002% of the optimum, and its layouts are visually indistinguishable from the optimal ones (Figure 4). At equal time it beats BSP-OT on most inputs: the fast preset (0.77 s) is closer to the optimum than BSP-OT with 16 plans (0.71 s) on eight of ten distributions, and the default (2.3 s) than 64 plans (2.9 s) on nine; on uniform points they are at 2.6% and 0.84%, against 45% and 32%. RasterFairy is 17–425% above the optimum on seven. Displacement is not the whole story, though: on the real embeddings BSP-OT stays within a few percent of the optimal displacement yet halves the fraction of grid neighbours that were input neighbours, because it scrambles local order.

ShuffleSnap is a local search, and its failures follow from that. Where it misses the optimum, the difference forms a few long closed loops in which every point is shifted by about one cell. Such a loop is optimal inside every window that sees only part of it, and a strided window cannot shift a point by one cell. We built inputs on which the method never moves while the optimum costs zero, so its relative excess is unbounded. On uniform points the loops grow with N: the excess rises from 0.1% at 10^3 points to 0.84% at 10^5 and 1.3% at 10^6. Concave masks and grids with a few surplus cells are harder: with uniform points, 1.4% on a blocky S-shaped mask (BSP-OT: 9.3%) and 3.1% with less than a row of surplus cells. Starting from a recursive bisection makes results worse.

The exact solver took a median of about two minutes at 10^5 points. When a proven optimum is worth that wait, it is the better choice; ShuffleSnap remains the fast answer, a warm start for the exact solver, and the method for larger sizes and tighter budgets. The code, raw results and figure scripts are released in two repositories.

References

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A Real embeddings

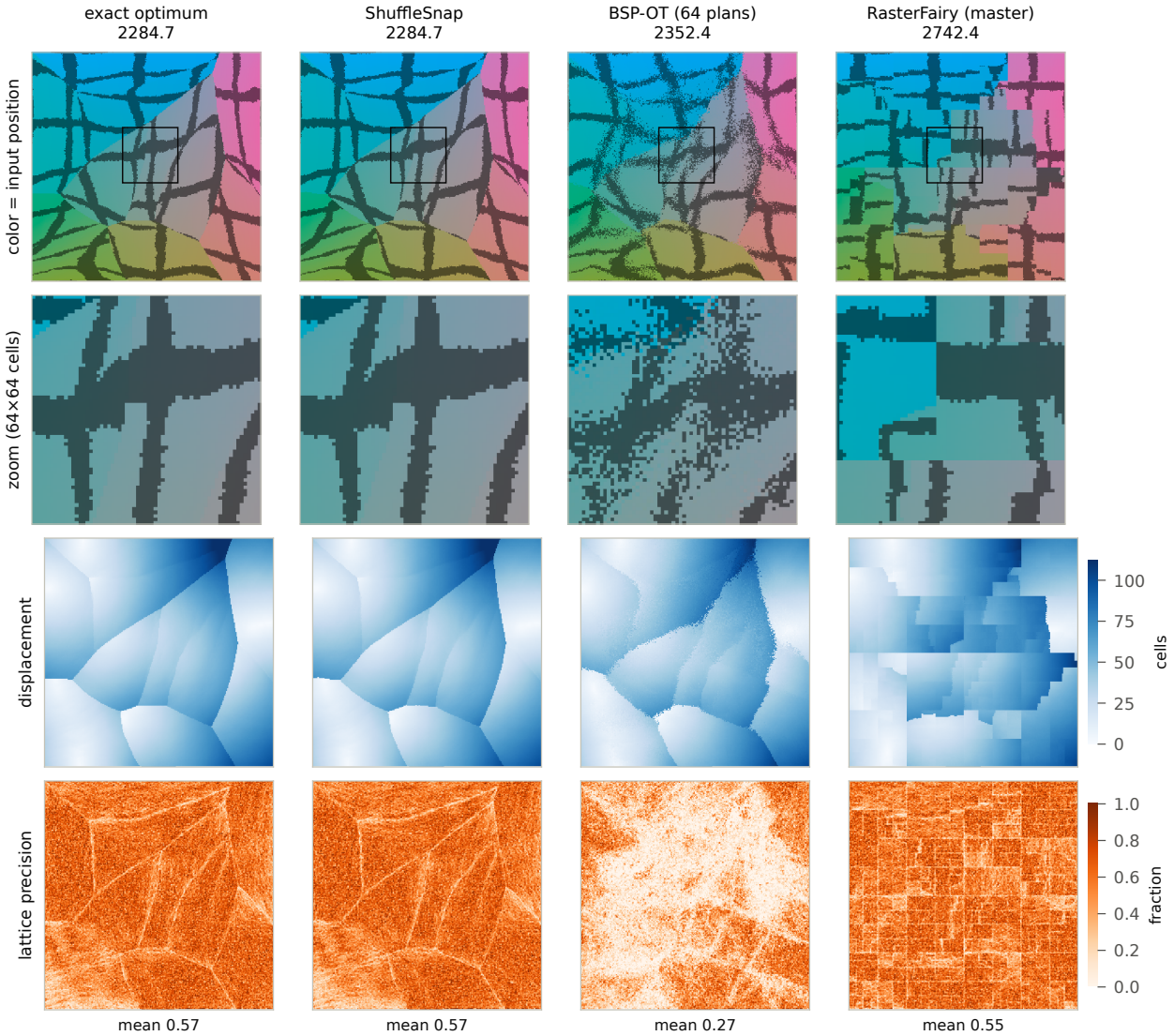


Figure 4: All 70,000 MNIST digits, UMAP layout snapped to a 265×265 grid. Colors and graph-paper lines encode input position as in Figure 2; numbers are the mean squared displacement in cells^2. Rows: the full grid, the 64×64 box enlarged, each point's displacement, and lattice precision (the fraction of a point's occupied 8-neighbour cells holding one of its 8 nearest input neighbours). BSP-OT stays within 3% of the optimal displacement but halves lattice precision; RasterFairy breaks the lines at rectangular seams.